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Joint Framework of Curriculum Learning and Knowledge Distillation for Noise-robust and Small-footprint Keyword Spotting

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ABSTRACT Spoken keyword spotting has been widely used to facilitate an always-on voice interface in consumer electronics, owing to its simplicity and low latency. Small-footprint keyword spotting based on tiny convolutional neural networks can be implemented on resource-constrained but energy-efficient microcontrollers in real time. However, it is difficult for tiny neural networks to learn the noise-robustness properties essential for successful voice interfaces. To overcome this problem, this study proposes a joint framework of curriculum learning and knowledge distillation for noise-robust small-footprint keyword spotting. The proposed joint framework applies noise mixture curriculum learning to a network that is sufficiently large for learning various noise situations. Subsequently, knowledge distillation is applied to compress the large network into a network small enough to be onboard microcontrollers. To enhance the effectiveness of the joint framework, we propose curriculum learning using a new noise mixture strategy and knowledge distillation with an effective ensemble of neural network snapshots for each curriculum stage. These proposed methods enable the effective learning of noisy situations by large networks and the transfer of noise robustness to small networks. The effectiveness of our joint framework is illustrated using the Google Speech Commands dataset with noise mixtures from various public noise datasets. Our joint framework results in superior performance in noisy situations compared with state-of-the-art noise-robust keyword spotting methods. Therefore, the proposed framework significantly improves the usability of voice interfaces in consumer electronics.

INDEX TERMS curriculum learning, data augmentation, joint framework, knowledge distillation, neural network compression, noise-robust keyword spotting, small-footprint keyword spotting

I. INTRODUCTION

With the widespread use of intelligent consumer electronics in daily life, voice interfaces have become widely used for natural and intuitive interactions. Spoken keyword spotting is widely used in consumer electronics to enable an always-on voice interface owing to its simplicity and low latency. Deep neural networks are commonly used for keyword spotting because of their superior accuracy [1]. A neural network trained on spoken keyword samples for identifying predefined keywords recognizes the trained keywords while distinguishing them from all the other words or phrases in an

audio stream. A keyword can involve a specific word or multiword phrase functioning as a wake-up word or command. Wake-up words can be “OK Google,” “Alexa,” and “Hey Siri” for artificial intelligence assistance. Moreover, keyword spotting is used to recognize simple commands, such as “Turn on,” “Turn off,” and “Play,” in various consumer electronics.

While other acoustic models, such as automatic speech recognition, are based on large neural networks with huge computational overheads, keyword spotting can be implemented with relatively small neural networks. Tiny

convolutional neural networks for keyword spotting have been actively studied to improve the performance of keyword spotting while reducing the computational load [2], [3], [4]. These networks convert the input audio into image-like spectrograms and process them efficiently using only a few network parameters. Additionally, studies have been actively conducted to propose various model compression techniques for deep neural networks [5], [6] [7]. One of effective techniques is knowledge distillation, which transfers the superior performance of a large teacher network to a small student network [8]. In the previous studies, keyword spotting models based on tiny convolutional networks was implemented on resource-constrained but energy-efficient microcontrollers in real time [9], [10].

The noise robustness of acoustic models is crucial for a successful voice interface. The performance degradation of the acoustic model due to ambient noise greatly damages the usability of the voice interface. Speech enhancement [11], [12] and data augmentation approaches have been proposed for the noise robustness of acoustic models. A promising data augmentation approach for improving noise robustness is the noise mixture training, in which background noise is injected to clean samples in the training data [13], [14]. Noise mixture training artificially introduces difficult training samples into the keyword spotting model, rendering the model more robust against noise. As it is difficult for neural networks to learn loud noise situations, curriculum learning for acoustic models has been studied [15], [16]. This is a learning method that increases the difficulty of samples—training easy samples such as clean samples first, and then difficult samples with loud noises, similar to how a person studies according to the curriculum [17]. Curriculum learning is widely used for noise-robust keyword spotting [18].

Recently, converging voice interfaces with wearable devices such as hearables and IoT devices for the smart home have been actively conducted [19], [20]. These tiny consumer electronics utilize spoken keyword spotting for a simple but valuable hand-free voice interface. Unlike smart speakers with high-performance processors [21], wearables and IoT devices utilize low-performance and low-power microcontrollers [22] for spoken keyword spotting. Although smart speakers show good performance using multiple microphones [23], it is difficult to mount multiple microphones due to the nature of those small devices [24]. Therefore, developing a noise-robust and small-footprint keyword spotting for the devices with minimal resources and a single microphone is required.

Although the effectiveness of curriculum learning has been demonstrated in keyword spotting, it is challenging to develop noise-robust and small-footprint keyword spotting because it is difficult to learn loud noise situations well [25], [26], [27]. When there are several difficult samples, such as loud noise situations, the joint framework of robust training and compression is a popular approach in various other domains. Existing studies in the fields of computer vision

and natural language processing have proposed joint frameworks for combining curriculum learning and knowledge distillation methods [28], [29]. However, studies on joint optimization methods for noise robustness are scarce in the field of acoustic modeling.

In this article, we propose a joint framework of curriculum learning and knowledge distillation for noise-robust and small-footprint keyword spotting. The joint framework comprises a two-phase algorithm. First, curriculum learning is applied to a sufficiently large network to learn various noise situations. Subsequently, knowledge distillation is applied to compress the large network into a network small enough to be onboard microcontrollers. Our main finding is that distilling the small network after applying curriculum learning to the large teacher network is superior to applying curriculum learning directly to the small network. To enhance the effectiveness of the joint framework, we propose curriculum learning using a new noise mixture strategy and knowledge distillation with an effective ensemble of neural network snapshots for each curriculum stage. Using these methods, large networks can effectively learn how to handle loud noise situations, and the resulting noise robustness can be transferred to smaller networks. The main contributions of this study are summarized as follows:

- 1) To the best of our knowledge, this is the first study on a joint framework of curriculum learning and knowledge distillation for noise-robust small-footprint keyword spotting.
- 2) We propose novel curriculum learning and knowledge distillation methods to enhance the effectiveness of the joint framework.
- 3) The proposed framework exhibits a state-of-the-art performance on four well-known noise datasets compared with current state-of-the-art methods in terms of noise robustness.

The remainder of this paper is organized as follows: In Section II, we review the related studies on small-footprint keyword spotting and noise-robust keyword spotting. In Section III, we describe the proposed joint framework for noise-robust small-footprint keyword spotting. In Section IV, we discuss the experimental results obtained using public datasets. Finally, we discuss our conclusions and future work in Section V.

II. RELATED WORK

A. SMALL-FOOTPRINT KEYWORD SPOTTING

With the advance of keyword spotting in both industry and academia, smart speakers and smart consumer electronics with voice interfaces are widespread. These devices utilize high-performance processors and multiple microphones for accuracy in various conditions, such as noise. Google Smart Speaker series have a quad-core application processor and over two microphones [21]. For such smart speakers, Yu et al. proposed a noise-robust keyword spotting model with noise cancellation using six microphones [23]. The number of parameters for the model is over 5.1 M.

Recently, wearables and IoT devices with always-on voice interfaces are more attractive for hands-free control and audio scene understanding [30]. For example, hearables can take advantage of built-in spoken keyword spotting for hands-free playback and volume control. In a smart home, IoT devices can be controlled with natural voice commands. These devices utilize low-power microcontrollers with only hundreds of KB of flash memory and tens of KB of RAM. Due to the nature of those small devices, it is difficult to mount multiple microphones. Fernandez-Marqueset et al. investigated tiny keyword spotting models for various microcontrollers [22]. Kim proposed a dedicated chip for wearables and IoT devices to use keyword spotting models. [24]. However, these studies on small-footprint keyword spotting do not consider noise robustness that is crucial for a successful voice interface.

Tiny convolutional neural networks have been widely used for small-footprint keyword spotting because they can efficiently learn spatial information from audio sequences. State-of-the-art networks use efficient structures composed of repeated blocks based on residual and depth-wise separable convolutions [2]. These convolutional networks use one-dimensional (1D) or partial two-dimensional (2D) convolution for efficiency. By utilizing temporal 1D convolutions, TC-ResNet requires fewer computations than the 2D approaches [3]. BC-ResNet employs broadcast residual learning to address the inefficiency of 2D convolution and the inferior performance of 1D convolution [4]. The authors applied frequency-wise 1D convolution to 2D audio features and then averaged the 2D features by frequency to obtain the temporal features. After temporal operations, the 2D features were subjected to residual mapping by broadcasting the 1D residual information.

Neural network compression has also received attention for small-footprint keyword spotting. It can be categorized into three approaches: pruning, quantization, and knowledge distillation. The pruning method reduces the model size and number of operations by eliminating unimportant parameters that do not degrade performance [31]. The quantization method approximates floating-point values by using a set of integers and scaling factors to achieve a smaller size and more efficient computations at the expense of a lower bit-width representation [32]. Knowledge distillation is a learning framework that utilizes a teacher–student network, where the teacher network transfers its knowledge to enhance the performance of the student network [8]. Ensemble distillation improves the performance of a distilled student by extracting knowledge from an ensemble of multiple teachers and encoding it in a student [33].

Song et al. proposed a knowledge distillation technique for a lightweight encoder as a replacement for complex speech frontends, such as mel-frequency cepstral coefficient (MFCC) or convolutions, while maintaining the performance of small-footprint keyword spotting [5]. Tucker et al. investigated two methods to improve the efficiency of a

small-footprint keyword spotting model: knowledge distillation and quantization [6]. Kim et al. introduced a method that simultaneously applies pruning, quantization, and knowledge distillation to small-footprint keyword spotting [7]. However, these studies on small-footprint keyword spotting do not consider noise robustness.

B. NOISE-ROBUST KEYWORD SPOTTING

Data augmentation has been widely applied to acoustic models to improve the neural network performance by adding extra diversity into the training data. For a successful voice interface, it is essential for the acoustic models to distinguish speech from background noise effectively. A promising augmentation approach for improving noise robustness is noise mixture training, in which environmental noise is added to clean samples in the training data [13], [14]. Noise mixture training introduces difficult training samples into models, making the models more robust against noise. Trinh et al. proposed ImportantAug, which augments the training data by injecting noise to unimportant speech regions. A data augmentation agent, trained to optimize the noise addition while minimizing its effect on performance, predicts the importance of speech [26].

Noise mixture training is a promising technique to achieve noise robustness for keyword spotting. However, it is difficult for a small-footprint keyword spotting model to learn loud noise situations, so the model becomes incompetent when faced with louder noise. Curriculum learning, which is progressive training, has been studied to overcome this problem for acoustic models [15], [16]. This method divides the learning process into several stages with gradually increasing difficulty of training samples: training with easy samples such as clean speech first, and then with difficult samples with loud noise [17]. Curriculum learning is more effective than conventional training in obtaining noise robustness, similar to how a person studies according to the curriculum. Therefore, research on keyword spotting widely applies curriculum learning for noise robustness.

As shown in Table I, although previous works for small-footprint keyword spotting improve noise robustness, its performance still needs to be improved when the noise is intense. Larger model or multi-microphone approaches show

TABLE I
RELATED WORK FOR NOISE-ROBUST KEYWORD SPOTTING

	Model Params #	Mic. #	Curriculum Learning	Knowledge Distillation	Public Dataset	Test SNR (dB)	Test Accuracy (%)
Larger model [26]	24000k	1	✓	✓	O	0 ~ 20	91.9
Multi-microphone [23]	5100k	6			X	-12 ~ 6	94.0
ImportantAug, 2022 [26]	100k	1			O	-12.5	56.5
[18]	-	1	✓		X	-	-
ConvMixer, 2022 [25]	119k	1	✓		O	-10	71.9
PKD, 2022 [27]	102k	1		✓	O	0	82.3
Proposed Joint Framework	27k	1	✓	✓	O	-12.5	78.6

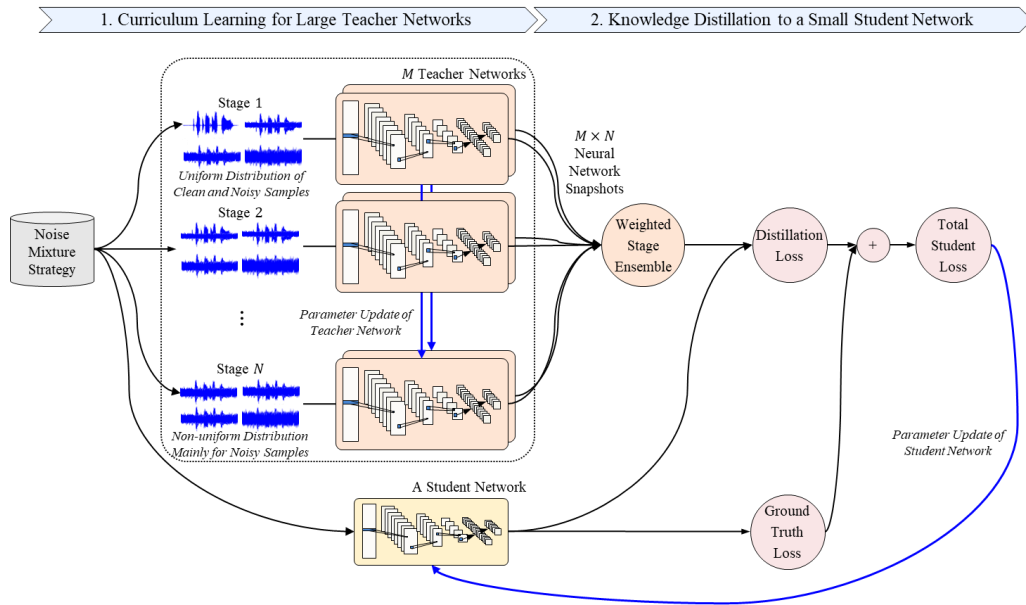


FIGURE 1. Overview of the proposed joint framework of curriculum learning and knowledge distillation. The proposed joint framework is characterized by applying the proposed noise mixture curriculum learning to a large model and compressing the large model into a small one with weighted ensemble knowledge distillation.

better robustness but are unsuitable for wearables and IoT devices [23], [26]. Developing robust and small keyword spotting is challenging because it is difficult for tiny neural networks to learn noise robustness. To overcome this, a joint framework of curriculum learning and compression techniques, such as knowledge distillation, is required to improve noise robustness for small-footprint keyword spotting. Although Guimaraes et al. utilized both curriculum learning and knowledge distillation for keyword spotting, they used distillation to exploit a huge self-supervised model, not to compress the keyword spotting model [26].

Curriculum learning or knowledge distillation is widely used for small-footprint keyword spotting. Higuchi et al. proposed dynamic curriculum learning to train a keyword spotting model on clean and noisy samples according to the data parameters [18]. These parameters automatically learn the difficulty of the classes and instances by gradient optimization. Ng et al. proposed a keyword spotting model called ConvMixer, with a traditional curriculum learning method to enhance noise robustness [25]. The training process is divided into five progressively harder stages and initially conditioned the model on clean speech samples without noise. A knowledge distillation method for noise-robust small-footprint keyword spotting called prototypical knowledge distillation (PKD), has been proposed to address the issue of network bias toward samples that are easy to train [27]. To train hard samples effectively, the PKD method utilizes a prototype distribution based on the distance between the class centroids and each embedding vector for knowledge distillation. However, more than research on jointly utilizing curriculum learning and knowledge distillation is required for small-footprint keyword spotting.

Studies on joint frameworks have been actively conducted in other fields, such as computer vision and natural language processing. Zhu et al. proposed a combination of curriculum learning and knowledge distillation methods to solve long-sentence training problems in natural language processing [27]. Panagiotatos et al. proposed a combination of curriculum learning and knowledge distillation methods for computer vision [28]. However, as such studies are scarce in the field of acoustic modeling, joint frameworks for noise robustness need to be studied.

III. PROPOSED JOINT FRAMEWORK

To overcome the difficulty in learning the noise robustness property in small-footprint keyword spotting, we proposed a joint framework that first learns a large teacher network that is robust to loud noise and compresses it into a small student network. To maximize the effectiveness of the joint framework, we proposed and combined novel curriculum learning and ensemble distillation methods. First, by applying the proposed noise mixture curriculum learning to a large model, the noise robustness in loud noise situations is maximized. Subsequently, the noise-robust large network is effectively compressed into a small network using the proposed ensemble distillation.

Fig. 1 shows the characteristics of the proposed joint framework. Unlike traditional curriculum learning, in which the range of noise strength is gradual widened, the proposed curriculum adopts a strategy of gradually increasing the ratio of noisy samples while learning noisy samples from an early stage for the effective learning. This demonstrates better performance for loud noise situations. The proposed ensemble distillation method utilizes neural network snapshots at each

stage of the curriculum learning process. Thus, the knowledge of the large teacher network is effectively transferred to the small student network, and the performance of the compressed network is more effectively preserved.

The proposed curriculum learning and knowledge distillation work together as follows: the proposed curriculum learning induces better performance in loud noise situations even though the performance when the noise is weak is slightly lower. Furthermore, such performance degradation in weak noise situations is mitigated by proposed knowledge distillation which ensembles all neural network snapshots generated during curriculum learning. Therefore, a distilled network gets both good performance at weak noise from early-stage network snapshots and good performance at loud noise from later-stage network snapshots.

A. PROPOSED CURRICULUM LEARNING

Unlike the gradual widening of the loudness range of noisy samples in traditional curriculum learning, we proposed a curriculum learning of training clean and noisy samples together from the initial stage but gradually increasing the proportion of noisy samples in each subsequent stage. As a result of the proposed curriculum learning, the performance of the neural network trained up to the initial stage improves under weak noise situations, and the performance of the neural network trained up to the final stage improves under loud noise situations. These neural network snapshots efficiently provide information on various noise intensities when performing knowledge distillation later.

The effective noise mixture strategy for each curriculum stage is the key to successful curriculum learning for noise-robust keyword spotting. In the curriculum learning process, the noise mixture strategy determines how to mix noise to clean speech at each stage to create training samples for training a neural network. The noise mixture strategy derives the characteristics of the final neural network as the stage progresses. For example, traditional curriculum training that gradually widens the SNR range of noise eventually leads to

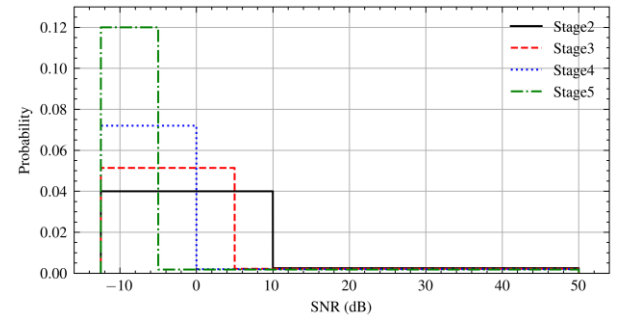


FIGURE 2. SNR distribution R_n for each stage n . The probability of selecting low-SNR samples for each subsequent stage is increased.

neural networks performing well in various noise situations. However, traditional curriculum learning is less effective because it is difficult for tiny neural networks to learn loud noise situations. Therefore, the proposed curriculum learning focuses more on the loud noise situation as the stage progresses, improving the performance in loud noise even if the performance drops slightly when the noise is weak.

To enhance noise robustness, the signal-to-noise ratio (SNR) was employed as the main component of the noise mixture strategy. The SNR of an audio sample is defined as the ratio of the power of speech to the power of the background noise, as shown in (1).

$$SNR = 10 \log_{10} \left(\frac{P_{speech}}{P_{noise}} \right), \quad (1)$$

where P denotes the average power. An SNR of less than 0 dB indicates that the background noise is louder than the speech. An SNR of -10 dB indicates that the power of the noise is 10 times greater than that of the speech in a sample.

Algorithm 1 shows the training process of the proposed curriculum learning. The training process was divided into the N progressively more difficult stages. For each stage, the teacher network is trained with noise mixed dataset D_{Mix} as much as e epoch. The samples of the dataset D_{Mix} are newly synthesized for each epoch according to the SNR distribution R_n of the corresponding stage. A clean sample d_{speech} are mixed with a randomly selected noise d_{noise} in SNR r . The parameters θ^T of the teacher network f trained with D_{Mix} . As a result, N neural network snapshots θ^{T_n} of the teacher network are obtained.

The SNR distribution R_n consists of the sampling and main ranges. The SNR for noise samples is selected from the main range with probability ρ and from the sampling range excluding the main range with probability $1 - \rho$. In the first stage, the SNR is uniformly selected from the sampling range since the sampling and main ranges are the same. In the following stages, the main range becomes narrowed gradually to increase the proportion of the noisy samples. As a result, SNR distribution R_n gradually focuses on loud noise situations as the stages progress. The selected SNR distribution R_n for each stage n is shown in Fig. 2.

Algorithm 1 Proposed curriculum learning

Input: Speech dataset D_{Speech} , Noise dataset D_{Noise} , SNR distribution R_n

Output: N Teacher Network Snapshots

```

01: For  $n$  in 1:  $N$  do
02:   For  $e$  in 1:  $E_n$  do
03:     For  $d_{Speech}$  in  $D_{Speech}$  do
04:       Randomly select a  $d_{Noise}$  from  $D_{Noise}$ 
05:       Select a SNR  $r$  in SNR distribution  $R_n$ 
06:       Calculate weight factor  $\omega$  to get SNR  $r$ 
07:        $d_{Mix} \leftarrow d_{Speech} + \omega d_{Noise}$ 
08:     End for
09:     Train teacher network  $\theta^T$  with  $D_{Mix}$ 
10:   End for
11:   Get teacher network snapshot  $\theta^{T_n}$  for stage  $n$ 
12: End for

```

The settings of parameters for the proposed curriculum learning are summarized as follows:

R_n : R_n is a factor that adjusts to focus more on loud situations as the stage progresses. The sampling range has a lower bound of -15 dB for samples of the loudest noise and an upper bound of 50 dB for almost clean samples. The main ranges for each stage are [-15, 50] dB, [-15, 10] dB, [-15, 5] dB, [-15, 0] dB, and [-15, -5] dB, respectively. We selected ρ to 0.9. The larger ρ is, the higher the probability that the SNR of the main range is sampled. In other words, the larger ρ is, the higher the concentration of loud situations. We validated these choices in section IV.

N : The number of stages. We set the number of stages to 5, the same as other curriculum learning studies [18], [25].

In addition, speech augmentation, which is commonly used for training acoustic models, was applied under the same conditions in all stages [35].

Volume change: The volume of the clean samples was randomly varied in the range of 40%–180%.

Shifting: The speech in a sample was shifted by 25% to 125% to change the central position of the speech.

Resampling: Samples were randomly resampled in the range of 90%–110% to change the speed or pitch.

SpecAugment [36]: Samples were masked in the frequency and time domains in the range of 0 to 5.

B. PROPOSED ENSEMBLE DISTILLATION

We used knowledge distillation to compress the curriculum-based teacher networks into a smaller student network. We proposed an ensemble distillation method that considers noise robustness to maximize the effect of distilling a student network with curriculum-based teacher networks. The proposed ensemble distillation effectively uses all neural network snapshots for each curriculum stage.

The knowledge of teacher networks is reflected when calculating the loss function while distilling a student network. It is necessary to efficiently decide which network's knowledge to receive among many large networks. The proposed ensemble distillation, weighted stage ensemble distillation, adjusts the weight of which teacher's knowledge should be given priority according to training data when distilling students. Teachers, which are neural network snapshots of the previous curriculum learning phase, focus more on loud situations as the stages progress. Therefore, the more noisy the data to be trained, the more knowledge is passed on from the snapshots in the later stage, and the cleaner the data, the more knowledge is passed on from the snapshots in the early stage, as shown in Fig. 3.

We first explain how traditional knowledge distillation and traditional ensemble distillation work in terms of loss function. Then we describe ensemble distillation using all snapshots for each curriculum stage. Finally, we present the proposed weighted stage curriculum learning with loss function and weight adjustment method.

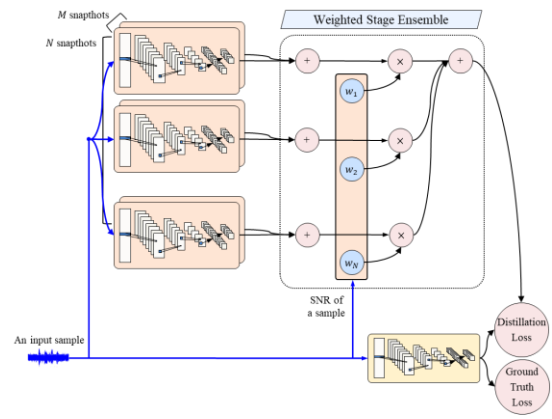


FIGURE 3. Overview of proposed ensemble distillation. With SNR of a sample, control which snapshots to transfer the most knowledge.

In traditional knowledge distillation, a teacher network is used for distilling a student network. The teacher logit output can be represented as $z^T = f(\theta^T, x)$, where x is the input sample, and θ^T refers to the trained parameters of the teacher network f . Moreover, $z^S = g(\theta^S, x)$ denotes the student logit output, where θ^S represents the parameters of the student network g . To determine the best parameters θ^S , the student is trained to minimize the cross entropy $\mathcal{H}$ between the ground truth y and the student probability output p^S after applying the SoftMax function σ , as shown in (2) and (3).

$$p^S = \sigma(z^S) = \frac{\exp(z_k^T)}{\sum_{j=1}^K \exp(z_j^T)} \text{ for all classes } k, \quad (2)$$

$$L_{CE}^S = \mathcal{H}(y, p^S) = \sum_{k=1}^K -y_k \log(p_k^S). \quad (3)$$

Along with the cross-entropy loss L_{CE}^S , the Kullback–Leibler (KL) divergence between the softened probability outputs of the teacher and student is considered to transfer the knowledge of the teacher network, as shown in (4) and (5). The softened probability outputs p^T and p^S are smoothed by a temperature τ , as in other distillation studies. Consequently, the total loss function is the weighted sum of the cross-entropy loss L_{CE}^S and the knowledge distillation loss L_{KD} with the loss weight λ , as shown in (6).

$$p^T = \sigma(z^T, \tau) = \frac{\exp(z_k^T/\tau)}{\sum_{j=1}^K \exp(z_j^T/\tau)} \text{ for all classes } k, \quad (4)$$

$$L_{KD} = \tau^2 KL(p^T, p^S) = \tau^2 \sum_{k=1}^K p_k^T \log(p_k^T/p_k^S), \quad (5)$$

$$L^S = (1 - \lambda)L_{CE}^S + \lambda L_{KD}. \quad (6)$$

In traditional ensemble distillation, multiple teacher networks are used to distill a student network. Let $z^{Tm} = f(\theta^{Tm}, x)$ denote the logit output of each teacher. Some independently trained neural networks with the same architecture f , same training algorithm on the same data distribution, and different initializations by random seeds result in teachers with different parameters θ^{Tm} . As shown in (7), the mean softened output P^E of N teachers improves the performance. Thus, the KL divergence between the mean

softened probability output P^E of the teachers and the probability output p^S of a student is used for ensemble distillation, as shown in (8). The ensemble distillation loss L_{ED} replaces the knowledge distillation loss L_{KD} in the total student loss, as shown in (9):

$$P^E = \sigma(\frac{1}{M} \sum_{m=1}^M z^{T_m}, \tau), \quad (7)$$

$$L_{ED} = \tau^2 KL(P^E, p^S), \quad (8)$$

$$\hat{L}^S = (1 - \lambda)L_{CE}^S + \lambda L_{ED}. \quad (9)$$

The proposed ensemble distillation method utilizes neural network snapshots learned up to the n -th stage to improve the efficiency of the ensemble. The network parameters $\theta^{T_{m,n}}$ of neural network snapshots can be obtained from as many as n curriculum stages for the m -th network parameter θ^{T_m} . Let $z^{T_{m,n}} = f(\theta^{T_{m,n}}, x)$ denote the logit output of each teacher. With N snapshots for each M independently trained neural networks, a total of $M \times N$ snapshots were used by the ensemble. When assembling a large number of networks, it is important to induce different networks to create synergy without conflict. Simply ensemble them all, as in (10), can often cause conflict.

$$\dot{P}^E = \sigma(\frac{1}{M \times N} \sum_{m=1}^M \sum_{n=1}^N z^{T_{m,n}}, \tau), \quad (10)$$

In the proposed ensemble distillation, the ensemble differs according to the training samples used to train the neural network snapshots $\theta^{T_{m,n}}$. Unlike all the snapshots contributing equally to the ensemble, the proposed strategy is to vary the contribution of snapshots to the ensemble according to the training sample. The proposed strategy for adjusting the weight w_n according to the training sample is shown in algorithm 2. The weight w_n emphasizes the snapshots for the n -th curriculum stage, as shown in (11).

$$\hat{P}^E = \sigma(\frac{1}{M \times N} \sum_{m=1}^M w_n \sum_{n=1}^N z^{T_{m,n}}, \tau), \quad (11)$$

The $\sum_{n=1}^N z^{T_{m,n}}$ denotes the subset ensemble with only snapshots at the n -th stage. Thus, the subset ensemble is a

subset of snapshots $\theta^{T_{m,n}}$. If the SNR of a training sample x is included in the main range of the SNR distribution R_n at the n -th stage, the weight w_n should be increased to emphasize the subset ensemble at the n -th stage. The weight w_s is assigned the constant α if the SNR is included in the main range of the SNR distribution R_n , and the constant β if not included, as shown in (12). The constant α is greater than the constant β , and these constants are set to appropriate values through a grid search. Consequently, the proposed ensemble distillation method adaptively emphasizes the snapshots for the n -th curriculum stage according to the training sample x .

$$w_n = \begin{cases} \alpha & \text{if } x \in \text{main range of } R_n \\ \beta & \text{else} \end{cases} \quad (12)$$

The settings of parameters for the proposed ensemble distillation are summarized as follows:

τ : The temperature τ of the knowledge distillation loss. The larger τ is, the smoother the teacher's logit is transmitted to the student. We set the temperature τ to 5 by the grid search, one of the commonly selected values for knowledge distillation.

λ : The loss weight λ of the knowledge distillation loss. The larger λ is, the larger the reflection ratio of distillation loss. We set the loss weight λ to 0.1 by the grid search, one of the commonly selected values for knowledge distillation.

N : The number of stages. We set the number of stages to 5.

M : The number of multiple teacher networks. We set the number of stages to 5. As M increases, the performance increases, but the learning cost also increases.

w_n : w_n is a factor that adjusts to focus on snapshots for loud situations. The w_n consists of α and β . As α increases, the weight of snapshots for loud situations increases. As β increases, snapshots are reflected evenly. We selected α to 1 and β to 0. We validated these choices in section IV.

IV. EXPERIMENTS

In this section, we present the experiments conducted to evaluate the effectiveness of the proposed joint framework for noise-robust and small-footprint keyword spotting. Our experiments examine keyword spotting models' performance and model size according to methods. To evaluate noise robustness, we compared the accuracy for each SNR level by adjusting the noise volume in the unseen noise environment setting. To compare the methods quantitatively, we use four public noise datasets: MUSAN [39], QUT [40], UrbanSound8K [41], and WHAM [42].

Our experiment consists of five parts. The proposed joint framework was compared to three state-of-the-art methods: ConvMixer, ImportantAug, and PKD. We verified the efficacy of the proposed curriculum learning compared to baseline curriculum learning methods: without curriculum learning and traditional curriculum learning. Moreover, we verified the efficacy of the proposed weighted stage ensemble distillation compared to baseline ensemble distillations: ensemble distillation without curriculum

Algorithm 2 Proposed ensemble distillation

Input: $M \times N$ Teacher Network Snapshots

Output: A Student Network

```

01: For  $e$  in 1:  $E_n$  do
02:   For  $d_{Mix}$  in  $D_{Mix}$  do
03:     For  $n$  in 1:  $N$  do
04:        $w_n \leftarrow \beta$ 
05:       If SNR of  $d_{Mix}$  is main range of  $R_n$ 
06:          $w_n \leftarrow \alpha$ 
07:       End if
08:     End for
09:     Get  $\hat{L}^S$  and  $\hat{P}^E$  with  $w_n \sum_{n=1}^N z^{T_{m,n}}$ 
10:   End for
11:   Train student network with  $\hat{L}^S$ 
12: End for

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learning, ensemble distillation with curriculum learning, and stage ensemble distillation. Furthermore, we explored the hyperparameters of the proposed curriculum learning and knowledge distillation methods. Finally, we extended the SNR of the public datasets and evaluated the accuracy according to the SNR in loud noise situations.

A. EXPERIMENTAL SETTINGS

We describe the experimental setup in detail to ensure the reproducibility of this study. To evaluate the effectiveness of the proposed joint framework, we selected experimental settings, including neural network architecture, data preprocessing, and training hyperparameters.

We used BC-ResNet, a state-of-the-art neural network architecture, for small-footprint keyword spotting [4]. The BC-ResNet-8 architecture with 321k parameters was used as the teacher network, and the BC-ResNet-2 architecture with 27.3k parameters as the student network. To compose the input data of BC-ResNet, 1 s of raw audio was preprocessed into 2D features of size 49×40 by MFCC conversion. BC-ResNet uses these inputs to output classification results for eleven classes consisting of ten predefined keywords and one negative class. The training samples for the negative class included words other than the predefined keywords or background audio.

When training the teacher or student from scratch, the experimental settings were as follows: We implemented all the code in TensorFlow. We used the Adam optimizer with a default learning rate of 0.001 and a common mini-batch size of 256. When applying curriculum learning, the neural networks were trained for 4000 epochs, with 2000, 500, 500, 500, and 500 epochs in five stages. Even when curriculum learning was not applied, the neural networks were trained for 4000 epochs. As the number of training samples for the negative class was more than 10 times greater than that for one predefined keyword, the class weight of the negative class was set to 0.1 to prevent bias against the negative class.

The experimental settings specified when distilling a student from the teacher were as follows. Similar to the scratch learning, we used the Adam optimizer with a learning rate of 0.001 and a mini-batch size of 256. The students were trained for 4000 epochs, regardless of the distillation method.

B. DATASETS

We used the Google Speech Commands (GSC) dataset version 2 in all our experiments [37]. This dataset included approximately 100k single-word audio of 35 unique words. The neural networks were trained using the training set and evaluated using the test set. Randomly selected noise and clean samples were mixed at different SNR levels to set up the noise environments. We used the FSK50k [38] dataset for training and the MUSAN [39], QUT [40], UrbanSound8K [41], and WHAM [42] datasets to evaluate the performance in unseen noise environments.

MUSAN: We employed the public GSC-MUSAN dataset used in a related work for noise-robust keyword spotting [26], mixed with noise from the MUSAN dataset, to augment clean samples from the GSC dataset. As the recordings in MUSAN have variable lengths, GSC-MUSAN only used the initial 1 s of each recording and discarded recordings that were shorter than that, as the samples were limited to a maximum of 1 s. After removing the short samples, GSC-MUSAN randomly selected 175 noise audio files and combined them with the samples from the test set.

QUT: This dataset contains an audio file named “HOME-LIVINGB-1.wav,” which comprises 40 min of background noise recorded in a living room setting. The audio file was used to construct the test data in a manner similar to that of a related work for noise-robust keyword spotting [26]. We randomly selected segments from this noise audio file and blended them with the clean samples from the test set. Subsequently, we modified the file sampling rate from 48 kHz to 16 kHz to match the GSC dataset.

UrbanSound8K: This dataset contains 8732 audio files of urban sounds shorter than 4 s from 10 classes: air conditioner, car horn, children playing, dog bark, drilling, engine idling, gunshot, jackhammer, siren, and street music. All the excerpts were obtained from field recordings uploaded at www.freesound.org. This dataset was also used in a manner similar to the QUT dataset.

WHAM: The WHAM dataset, or WSJ0 Hipster Ambient Mixtures, is a compilation of two-speaker mixtures derived from the wsj0-2 mix dataset, merged with authentic ambient noise samples. These samples were gathered from various public spaces such as coffee shops, restaurants, and bars within the San Francisco Bay Area. This dataset was also used in a manner similar to the QUT dataset.

C. STATE-OF-THE-ART METHODS

We evaluated the keyword spotting performance of our joint framework and compared it with that of state-of-the-art methods for noise-robust keyword spotting.

ConvMixer: Ng et al. [25] proposed ConvMixer, a new convolutional neural network architecture that adds a mixer layer that mixes the frequency and temporal domain features, along with 1D temporal, 2D frequency × temporary and partial 2D convolutions. ConvMixer applies traditional noise mixture curriculum learning as a training strategy to enhance the noise robustness. The training process was divided into five progressively more difficult stages. Initially, ConvMixer was trained on clean speech samples without noise. In the subsequent three stages, noise set was introduced to noisy samples in increments of -5 dB as follows: {clean, 0 dB}, {clean, 0 dB, -5 dB}, {clean, 0 dB, -5 dB, -10 dB}.

ImportantAug: Trinh et al. [26] proposed ImportantAug, a method for enhancing the quality of noise samples for training acoustic models. This approach involves injecting noise exclusively to unimportant speech areas while leaving important regions untouched. The significance level for each

TABLE II
COMPARISON WITH STATE-OF-THE-ART NOISE-ROBUST AND SMALL-FOOTPRINT KEYWORD SPOTTING METHODS

COMPARISON WITH STATE OF THE ART NOISE ROBUST AND SMALL FOOTPRINT NETWORK DENOISING METHODS																													
	Model Parms #	Accur- acy on Clean (%)	Accuracy on MUSAN (%)							Accuracy on QUT (%)							Model Parms #	Accur- acy on Clean (%)	Accuracy on UrbanSound8k (%)					Accuracy on WHAM (%)					
			40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB	40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB				20 dB	15 dB	10 dB	5 dB	0 dB	20 dB	15 dB	10 dB	5 dB	0 dB
Curriculum Learning	27k	93.1	93.1	92.9	92.3	90.7	86.1	74.5	70.7	93.0	92.9	92.0	89.2	80.1	56.8	49.6	Curriculum Learning	27k	93.1	92.5	91.7	90.3	87.7	82.5	92.8	92.1	91.0	88.6	84.0
ConvMixer, 2022 [25]	119k	93.2	-	-	90.8	-	83.0	71.9	-	-	-	-	-	-	-	-	PKD, 2022 [27]	102k	97.7	96.7	95.6	93.8	89.2	82.3	96.6	95.9	94.2	89.4	79.4
ImportantAug, 2022 [26]	100k	95.0	94.9	94.8	94.3	92.6	86.7	65.0	56.5	95.2	94.9	94.2	91.1	76.5	38.7	28.0													
Proposed Joint Framework	27k	97.1	97.0	96.8	96.4	95.2	91.1	82.3	78.6	97.3	97.0	96.6	94.3	87.8	68.5	64.1	Proposed	27k	97.1	96.4	95.8	94.9	93.1	89.2	96.9	96.4	96.1	94.5	91.1

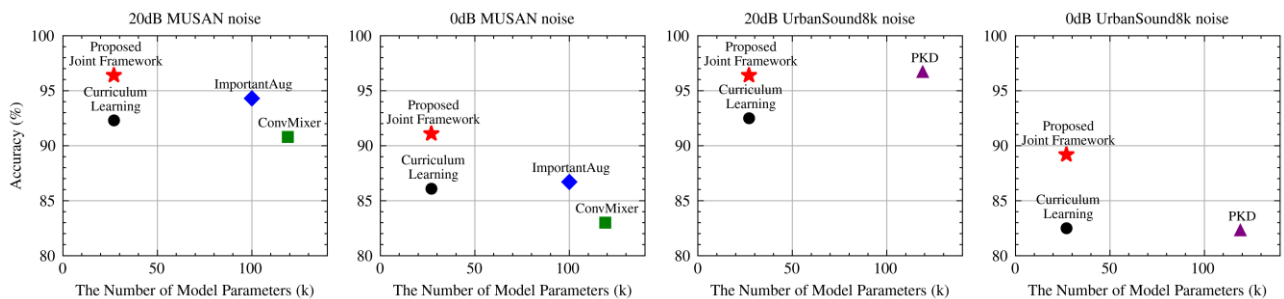


FIGURE 4. Efficacy of the proposed joint framework compared with curriculum learning and state-of-the-art methods on the MUSAN and UrbanSound8k datasets. The proposed joint framework showed accuracy improvement and model size reduction compared to the state-of-the-art.

speech was estimated by the augmentation agent trained to optimize the amount of noise introduced while minimizing the effect on accuracy.

PKD: Kim et al. [27] presented a robust knowledge distillation method and a feature extraction method with a multilayer operation. They proposed three different distance metrics for the knowledge distillation and a novel feature extraction method using the distribution between class centroids and embedding vectors. Background noise was randomly injected, and random time-shifting was applied.

D. EXPERIMENTAL RESULTS

1) PROPOSED JOINT FRAMEWORK RESULTS

Table II presents the results of the comparison between the proposed joint framework and recent state-of-the-art methods. The accuracies of the state-of-the-art ConvMixer, ImportantAug, and PKD methods were compared in terms of the accuracy and model size. The accuracy of keyword spotting was compared at different SNRs for four noise datasets: MUSAN, QUT, UrbanSound8k, and WHAM. The performance of ImportantAug on the MUSAN dataset was the experimental result under the observed noise condition, and the performances of the other methods, including the proposed method, were the experimental results under the unseen noise condition. The proposed joint framework showed better keyword spotting performance than all the state-of-the-art methods under all the noise conditions. In particular, the proposed joint framework showed significant performance improvement in loud noise conditions, where

the power of the noise audio was greater than or equal to that of the speech audio.

Fig. 4 illustrates the keyword spotting accuracy of the proposed joint framework compared with that of the existing curriculum learning method. The black point shows the results of applying curriculum learning directly to the small network, and the red star demonstrates the result for the proposed method of distilling the small network after applying curriculum learning to the large teacher network. The proposed joint framework showed an average 8% improvement in accuracy and an average four times reduction in model size compared to the state-of-the-art methods. The proposed joint framework showed an accuracy of 96.4% at 20 dB MUSAN, 91.1% at 0 dB MUSAN, 96.4% at 20 dB UrbanSound8k, and 89.2% at 0 dB UrbanSound8k, compared with the accuracies of the curriculum-learning-only of 92.3%, 86.1%, 92.5%, and 82.5%, respectively. A significant performance improvement was observed for loud noise samples when the joint framework was applied.

2) PROPOSED CURRICULUM LEARNING RESULTS

In this experiment, we demonstrate the performance of the proposed curriculum learning in our joint framework. We verified the efficacy of the proposed curriculum learning compared to baseline curriculum learning methods: without curriculum learning and traditional curriculum learning similar to one used in ConvMixer.

Table III shows the accuracy of keyword spotting on the MUSAN noise dataset resulting from the application of curriculum learning to a small network (BC-ResNet-2). “No

TABLE III
EVALUATION OF SMALL AND LARGE NETWORKS ACCORDING TO CURRICULUM METHODS

	Model Size (Param s #)	Accuracy (%) on Clean (F1-score)	Averaged Accuracy (%) of Noises (F1-score)	Accuracy (%) on MUSAN (F1-score)								Accuracy (%) on QUT (F1-score)								Accuracy (%) on UrbanSound8k (F1-score)						Accuracy (%) on WHAM (F1-score)					
				40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB	40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB	20 dB	15 dB	10 dB	5 dB	0 dB	20 dB	15 dB	10 dB	5 dB	0 dB				
No Curriculum	Small (27k)	95.7 (0.94)	86.5 (0.82)	95.6 (0.94)	95.4 (0.93)	94.8 (0.92)	93.1 (0.90)	87.1 (0.82)	73.0 (0.65)	68.9 (0.60)	95.5 (0.93)	95.3 (0.93)	94.5 (0.92)	92.1 (0.88)	81.5 (0.75)	53.2 (0.44)	45.8 (0.36)	94.6 (0.92)	93.7 (0.91)	92.1 (0.89)	88.9 (0.85)	82.8 (0.78)	95.0 (0.93)	94.5 (0.92)	93.1 (0.90)	90.4 (0.87)	84.8 (0.80)				
	Large (313k)	97.9 (0.97)	91.4 (0.88)	97.9 (0.97)	97.8 (0.96)	97.4 (0.96)	96.4 (0.94)	93.4 (0.90)	82.2 (0.75)	77.4 (0.69)	97.9 (0.97)	97.7 (0.96)	97.2 (0.95)	95.8 (0.93)	89.7 (0.85)	64.3 (0.55)	55.1 (0.45)	97.4 (0.96)	96.9 (0.95)	95.9 (0.94)	94.2 (0.91)	90.8 (0.86)	97.6 (0.96)	97.2 (0.95)	96.4 (0.94)	94.9 (0.92)	91.7 (0.88)				
Traditional Curriculum	Small (27k)	91.7 (0.89)	84.6 (0.80)	91.7 (0.91)	91.5 (0.92)	91.7 (0.87)	85.2 (0.73)	69.4 (0.64)	91.7 (0.91)	91.4 (0.90)	88.3 (0.80)	80.5 (0.58)	58.4 (0.49)	51.1 (0.41)	91.6 (0.90)	89.4 (0.86)	86.9 (0.81)	91.6 (0.88)	90.8 (0.87)	89.4 (0.86)	88.9 (0.83)	81.9 (0.77)	91.7 (0.89)	91.1 (0.88)	90.1 (0.86)	88.1 (0.84)	84.0 (0.79)				
	Large (313k)	98.0 (0.97)	91.3 (0.88)	98.0 (0.97)	97.8 (0.96)	97.5 (0.96)	96.5 (0.95)	93.6 (0.90)	82.0 (0.75)	77.4 (0.69)	98.0 (0.97)	97.8 (0.96)	97.4 (0.95)	95.9 (0.89)	89.7 (0.85)	62.9 (0.54)	53.7 (0.44)	97.4 (0.96)	96.8 (0.95)	95.9 (0.94)	94.2 (0.91)	90.5 (0.86)	97.6 (0.96)	97.2 (0.95)	96.5 (0.94)	95.0 (0.92)	90.9 (0.87)				
Proposed Curriculum	Small (27k)	93.1 (0.91)	85.3 (0.81)	93.1 (0.91)	92.9 (0.90)	92.3 (0.89)	90.7 (0.87)	86.1 (0.81)	74.5 (0.67)	70.7 (0.62)	93.0 (0.90)	92.9 (0.90)	92.0 (0.89)	89.2 (0.85)	80.1 (0.74)	56.8 (0.48)	49.6 (0.40)	92.5 (0.90)	91.7 (0.89)	90.3 (0.87)	87.7 (0.84)	82.5 (0.77)	92.8 (0.90)	92.1 (0.89)	91.0 (0.88)	88.6 (0.85)	84.0 (0.79)				
	Large (313k)	97.4 (0.96)	91.4 (0.88)	97.3 (0.96)	97.1 (0.96)	96.8 (0.96)	93.4 (0.94)	83.9 (0.79)	79.7 (0.72)	77.3 (0.72)	97.3 (0.97)	97.1 (0.95)	96.5 (0.94)	89.3 (0.84)	66.7 (0.58)	58.3 (0.49)	58.3 (0.49)	96.8 (0.96)	96.4 (0.94)	95.5 (0.94)	94.0 (0.91)	91.2 (0.87)	97.1 (0.95)	96.7 (0.95)	96.0 (0.94)	94.6 (0.92)	91.8 (0.88)				

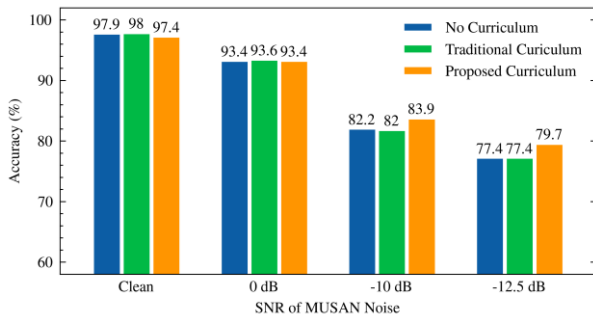


FIGURE 5. Comparison of the accuracy of the large networks between the curriculum methods on the MUSAN dataset

curriculum” denotes the result of applying a random noise mixture, such as ImportantAug and PKD. “Traditional curriculum” denotes the result of applying a general curriculum learning approach, such as ConvMixer. The experimental results showed that the proposed noise mixture curriculum exhibited the best performance.

Fig. 5 shows a comparison of keyword spotting accuracy according to the application of curriculum learning to a large network (BC-ResNet-8). The traditional noise mixture curriculum learning applied in ConvMixer had no performance improvement effect compared with the random noise mixture used in ImportantAug and PKD. When the proposed curriculum learning was applied, the performance was improved to 83.9% at -10 dB and 79.7% at -12.5 dB, compared with 82.2% and 77.4%, respectively, in loud noise situations. In addition, compared with the performance of the small network in Fig. 5, that is, 74.6% at -10 dB and 70.7% at -12.5 dB, a significant performance improvement was shown. Therefore, it is difficult to cope with loud noise with a small network, even when curriculum learning is applied.

Fig. 6 shows the improvement trend of keyword spotting accuracy at -12 dB on the MUSAN noise dataset by epochs as learning progressed according to the applied curriculum learning method. When a random noise mixture was applied without curriculum learning, the performance converged to 77.4% at 3000 epochs. In addition, when traditional

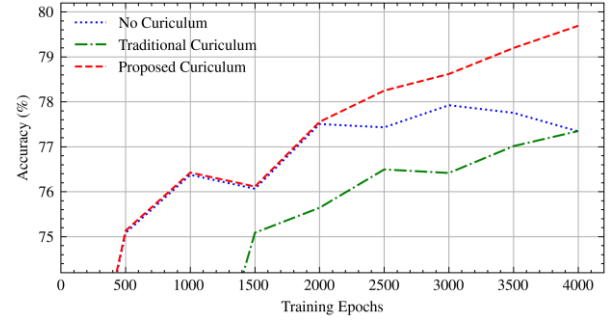


FIGURE 6. Accuracy of the large networks for training epochs according to curriculum methods

curriculum learning was applied, the performance improvement was slower than when traditional curriculum learning was not applied, and the highest performance was similar. When the proposed method was applied, the performance improved.

3) PROPOSED ENSEMBLE DISTILLATION RESULTS

In this experiment, we demonstrated the performance of the proposed ensemble distillation in our joint framework. We verified the efficacy of the proposed weighted stage ensemble distillation compared to three baseline ensemble distillations: ensemble distillation without curriculum learning, ensemble distillation with curriculum learning, and stage ensemble distillation.

Table IV shows the comparison results of keyword spotting accuracy according to the distillation methods. The performances of the teacher ensembles and distilled students were compared according to the distillation methods. Compared with learning from scratch using a small network without applying knowledge distillation, the performance was significantly improved when students were extracted through any distillation methods. In addition, ensemble distillation with proposed curriculum learning improved performance than without the curriculum learning. The comparison confirmed that the proposed weighted-stage ensemble distillation exhibited the best performance in all noise situations.

TABLE IV
EVALUATION OF TEACHER ENSEMBLE AND DISTILLED STUDENTS ACCORDING TO ENSEMBLE DISTILLATION METHODS

EVALUATION OF TEACHER ENSEMBLE AND DISTILLED STUDENT'S ACCORDING TO ENSEMBLE DISTILLATION METHODS																											
	Before or After of Distillation (Model Params #)	Accuracy (%) on Clean (F1-score)	Averaged Accuracy (%) of Noises (F1-score)	Accuracy (%) on MUSAN (F1-score)							Accuracy (%) on QUT (F1-score)							Accuracy (%) on UrbanSound8k (F1-score)					Accuracy (%) on WHAM (F1-score)				
				40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB	40 dB	30 dB	20 dB	10 dB	0 dB	-10 dB	-12.5 dB	20 dB	15 dB	10 dB	5 dB	0 dB	20 dB	15 dB	10 dB	5 dB	0 dB
Ensemble Distillation without Curriculum	Before (1.5M)	98.5 (0.98)	92.5 (0.89)	98.5 (0.98)	98.3 (0.97)	98.0 (0.97)	97.0 (0.95)	94.4 (0.91)	84.5 (0.78)	80.1 (0.72)	98.4 (0.97)	98.2 (0.97)	97.8 (0.96)	96.7 (0.95)	91.5 (0.87)	67.8 (0.58)	59.2 (0.48)	97.9 (0.97)	97.3 (0.96)	96.4 (0.94)	95.1 (0.92)	91.8 (0.88)	98.0 (0.97)	97.7 (0.96)	97.1 (0.95)	95.7 (0.93)	92.9 (0.89)
	After (27k)	97.1 (0.95)	90.4 (0.85)	97.0 (0.95)	96.8 (0.94)	96.3 (0.92)	95.1 (0.85)	90.9 (0.85)	79.6 (0.69)	76.1 (0.64)	97.3 (0.95)	97.0 (0.95)	96.6 (0.94)	94.5 (0.91)	87.2 (0.80)	66.3 (0.49)	60.2 (0.41)	96.3 (0.94)	95.7 (0.93)	94.3 (0.91)	92.2 (0.88)	87.9 (0.82)	96.9 (0.95)	96.5 (0.94)	96.1 (0.92)	93.2 (0.89)	89.3 (0.83)
Ensemble Distillation with Curriculum	Before (1.5M)	98.1 (0.97)	92.8 (0.90)	98.0 (0.97)	97.9 (0.97)	97.6 (0.96)	96.8 (0.95)	94.5 (0.92)	86.0 (0.80)	82.1 (0.74)	98.1 (0.97)	97.9 (0.97)	97.3 (0.96)	96.1 (0.94)	91.3 (0.87)	70.7 (0.62)	62.5 (0.52)	97.5 (0.96)	97.1 (0.95)	96.3 (0.94)	95.2 (0.93)	92.6 (0.89)	97.8 (0.96)	97.3 (0.96)	96.7 (0.95)	95.5 (0.93)	93.2 (0.90)
	After (27k)	96.9 (0.95)	90.9 (0.85)	96.8 (0.94)	96.2 (0.94)	96.2 (0.92)	94.6 (0.85)	90.9 (0.85)	82.3 (0.72)	78.7 (0.66)	97.0 (0.95)	96.8 (0.95)	96.2 (0.94)	94.1 (0.91)	87.8 (0.81)	68.6 (0.50)	64.1 (0.43)	96.2 (0.94)	95.6 (0.93)	94.4 (0.91)	93.1 (0.88)	89.0 (0.83)	96.6 (0.94)	96.0 (0.93)	95.4 (0.92)	94.4 (0.90)	91.0 (0.85)
Stage Ensemble Distillation	Before (7.5M)	98.1 (0.97)	92.3 (0.89)	98.1 (0.97)	98.0 (0.97)	97.6 (0.96)	96.7 (0.95)	94.2 (0.91)	85.0 (0.78)	80.8 (0.72)	98.0 (0.97)	97.8 (0.96)	97.3 (0.96)	96.1 (0.94)	90.7 (0.86)	67.9 (0.58)	59.1 (0.48)	97.5 (0.96)	97.1 (0.95)	96.3 (0.94)	94.9 (0.92)	92.1 (0.88)	97.7 (0.96)	97.5 (0.96)	96.8 (0.95)	95.6 (0.93)	92.7 (0.89)
	After (27k)	97.0 (0.95)	90.5 (0.85)	97.0 (0.95)	96.8 (0.94)	96.2 (0.92)	94.6 (0.85)	90.7 (0.85)	80.7 (0.70)	76.9 (0.64)	97.0 (0.95)	96.8 (0.95)	96.2 (0.94)	94.1 (0.91)	87.0 (0.80)	67.9 (0.49)	62.7 (0.41)	96.2 (0.94)	95.6 (0.93)	94.4 (0.91)	92.3 (0.88)	88.3 (0.83)	96.6 (0.94)	96.0 (0.93)	95.4 (0.92)	93.6 (0.89)	90.0 (0.83)
Weighted Stage Ensemble Distillation	Before (7.5M)	98.4 (0.97)	92.9 (0.90)	98.3 (0.97)	98.2 (0.97)	97.9 (0.97)	96.9 (0.95)	94.8 (0.92)	85.8 (0.80)	82.0 (0.74)	98.3 (0.97)	98.1 (0.97)	97.7 (0.96)	96.6 (0.95)	91.7 (0.88)	70.3 (0.61)	61.9 (0.51)	97.8 (0.97)	97.3 (0.96)	96.5 (0.95)	95.3 (0.93)	92.6 (0.89)	98.0 (0.97)	97.6 (0.96)	97.0 (0.95)	95.8 (0.93)	93.3 (0.90)
	After (27k)	97.1 (0.95)	91.1 (0.86)	97.0 (0.95)	96.8 (0.95)	96.4 (0.94)	95.2 (0.92)	91.1 (0.86)	82.3 (0.72)	78.6 (0.66)	97.3 (0.97)	97.0 (0.95)	96.6 (0.94)	94.3 (0.91)	87.8 (0.81)	68.5 (0.50)	64.1 (0.43)	96.4 (0.94)	95.8 (0.93)	94.9 (0.92)	93.1 (0.89)	89.2 (0.84)	96.9 (0.95)	96.4 (0.94)	96.1 (0.92)	94.5 (0.90)	91.1 (0.85)

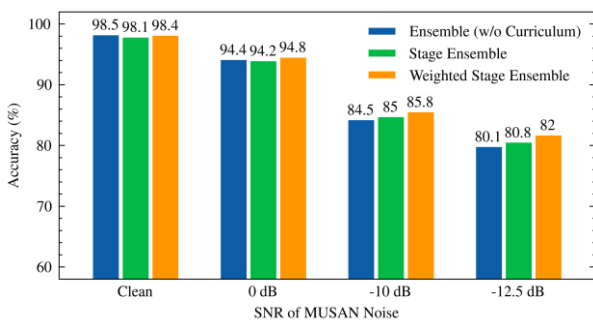


FIGURE 7. Ensemble accuracy of teacher networks according to ensemble methods on the MUSAN noise dataset

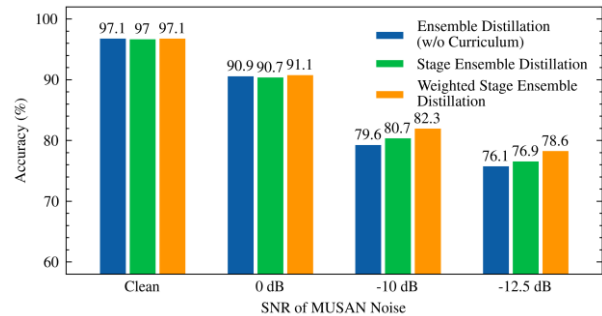


FIGURE 8. The accuracy comparison of the distilled students according to ensemble methods on the MUSAN noise dataset

Fig. 7 shows the keyword spotting accuracy on the MUSAN dataset according to the teacher ensemble methods. The ensemble using only the final teachers trained independently, as shown in (7), showed an accuracy of 80.1% at -12.5 dB. The stage ensemble, which uses not only the final teachers trained independently but also the intermediate result teachers for each stage, as shown in (10), showed an accuracy of 80.8% at -12.5 dB. Finally, the proposed weighted-stage ensemble, which provides weights to the intermediate result teachers for each stage according to the SNR of the training sample, as shown in (11), showed an accuracy of 82% at -12.5 dB. This was the best performance compared with the baseline.

Fig. 8 shows the distilled student networks accuracy on the MUSAN dataset according to the ensemble distillation methods. The ensemble distillation using only the final teachers trained independently without curriculum learning, as shown in (7), showed an accuracy of 76.1% at -12.5 dB. The stage ensemble, which uses not only the final teachers trained independently but also the intermediate result teachers for each stage, as shown in (10), showed an accuracy of 76.9% at -12.5 dB. Finally, the proposed weighted-stage ensemble, which provides weights to the

intermediate result teachers for each stage according to the SNR of the training sample, as shown in (11), showed an accuracy of 78.6% at -12.5 dB. This was the best performance compared with the baseline.

E. JOINT FRAMEWORK HYPERPARAMETERS

1) PROPOSED CURRICULUM LEARNING

As described in Section III, the proposed curriculum learning method varied the ratio of loud noise samples at each stage. Speech augmentation was simultaneously applied. In this experiment, we compared the keyword spotting accuracy according to the noise mixture strategy for each stage and whether speech augmentation was applied. We used MUSAN dataset for the comparisons.

Fig. 9 shows a comparison of the accuracy of a teacher network according to the probability ρ for the SNR distribution R_n . The selected probability ρ was 0.9. the accuracy improved slightly from 79.5% or 79.6% to 79.7% at -12.5 dB compared to the probability ρ was other values.

Fig. 10 shows a comparison of the accuracy of a teacher network according to the loud noise ratio for each stage. The selected curriculum increased the main range for each stage as follows: [-15, 50] dB, [-15, 10] dB, [-15, 5] dB, [-15, 0] dB,

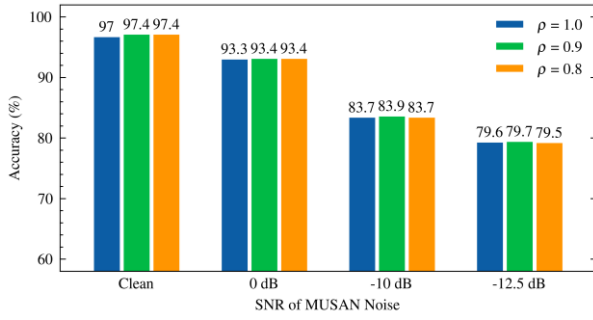


FIGURE 9. Comparison of the noise mixture strategy according to probability ρ of the main range on the MUSAN dataset

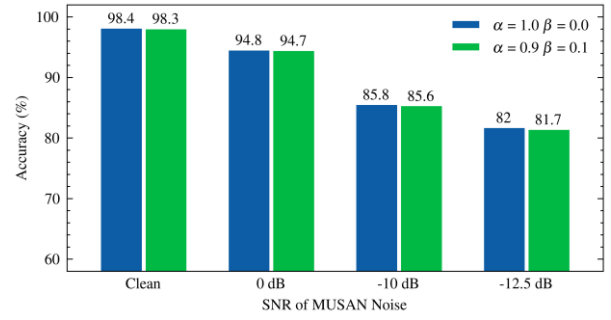


FIGURE 12. Comparison of the proposed ensemble according to α and β of the weight factor w_n on the MUSAN dataset

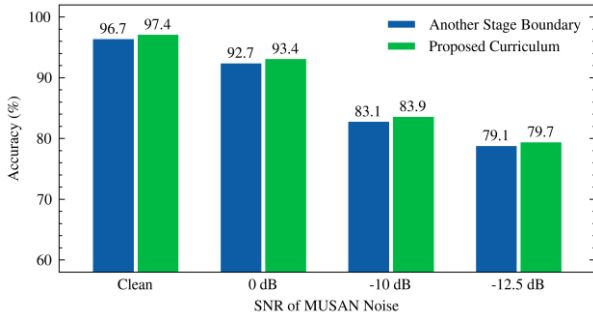


FIGURE 10. Comparison of the noise mixture strategy between a stage boundary on the MUSAN dataset

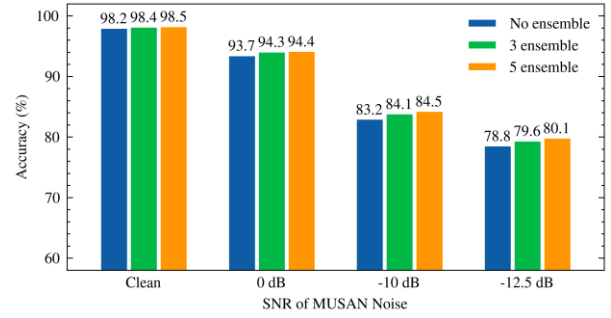


FIGURE 13. Comparison of the ensemble depending on the number of teachers on the MUSAN dataset

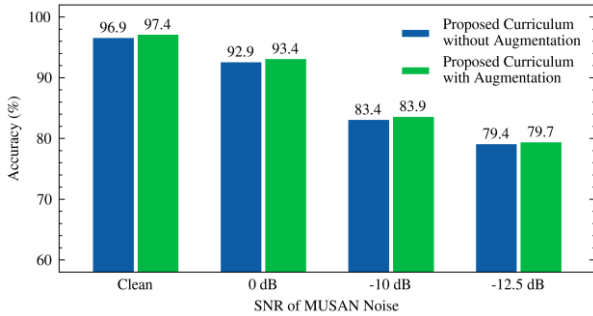


FIGURE 11. Comparison of the noise mixture strategy depending on whether augmentation was applied on the MUSAN dataset

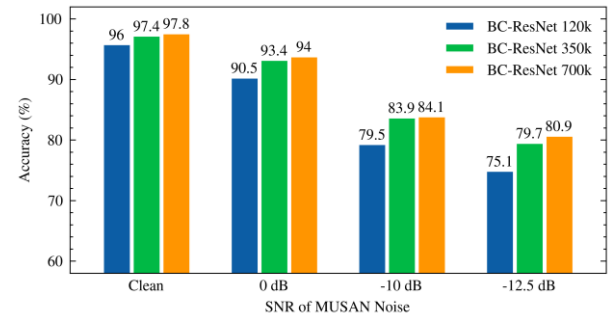


FIGURE 14. Comparison of the teacher according to the number of network parameters on the MUSAN dataset

and [-15, -5] dB, which showed better performance than that under other curriculums. In particular, the selected curriculum showed slightly better performance, 79.7% versus 79.1% at -12.5 dB, at the main ranges [-15, 50] dB, [-15, 5] dB, [-15, 0] dB, [-15, -5] dB, and [-15, -10] dB for each stage.

Fig. 11 shows a comparison of the accuracy of a teacher based on whether speech augmentation was applied. The accuracy improved slightly from 79.4% to 79.7% at -12.5 dB, when speech augmentation was applied.

2) PROPOSED ENSEMBLE DISTILLATION

As described in Section III, the proposed ensemble distillation method distills a student using neural network snapshots for each curriculum stage. In this experiment, we explored the hyperparameters for the proposed ensemble distillation,

including the number of teachers used and the size of a teacher network. We used MUSAN dataset for the exploration.

Fig. 12 shows a comparison of accuracy according to the constant α and β . The selected constant α and β were 1 and 0. The performance of the proposed ensemble, as shown in (11), with the constant α and β was better than that of the other choice. The accuracy improved slightly from 81.7% to 82% at -12.5 dB compared to the constant α and β were other value.

Fig. 13 shows a comparison of accuracy according to the number of teachers for the ensemble, as shown in (7). The performance of ensembles was better than that of a teacher. The accuracy improved as the number of teachers increased. The accuracy increased from 78.8% with no ensemble at -12.5 dB to 80.1% with an ensemble of five teachers.

Fig. 14 shows a comparison of the accuracy of a teacher network according to the number of network parameters. The larger the size of the teacher, the higher is the accuracy. In the case of a teacher with 120k parameters, the accuracy was 75.1%, whereas it was 79.7% when the number of parameters was increased to 350k and 80.9% when the number of parameters was increased to 700k.

The ensemble using a larger size and number of teachers shows better accuracy, but an appropriate size or number of teachers can be selected considering the limitations of computing resources.

F. RESULTS ON EXTENDED SNR OF NOISE DATSETS

So far, we conducted experiments using public datasets to compare three state-of-the-art methods. We extended the SNR of the public datasets and evaluated the accuracy according to the SNR in loud noise situations. Fig. 15 illustrates the keyword spotting accuracy of the proposed joint framework according to the SNR on four public noise datasets. The proposed joint framework showed an accuracy of 66.8% at -15 dB MUSAN, 54.1% at -15 dB QUT, 59.6% at -15 dB UrbanSound8k, and 53.3% at -15 dB WHAM, compared with the accuracies of the curriculum learning of 57.2%, 43.2%, 50.2%, and 43.9%, respectively. A significant performance improvement was observed for loud noise samples when the joint framework was applied.

V. CONCLUSIONS

This paper presented the first study on a joint framework of curriculum learning and knowledge distillation for noise-robust and small-footprint keyword spotting. Our main finding is that distilling the small network after applying curriculum learning to the large teacher network is superior to applying curriculum learning directly to the small network. Keyword spotting is a voice interface technology already widespread in consumer electronics such as smart speakers. However, the keyword spotting models must be more robust and lighter to apply to battery-powered and resource-constrained devices. We believe that the proposed joint framework will promote keyword spotting-based voice interfaces for wearable devices such as hearables and IoT devices for smart homes.

This study proposed a joint framework that benefits from curriculum learning and knowledge distillation. We proposed curriculum learning with a new noise mixture strategy and knowledge distillation with an effective ensemble of neural network snapshots for each curriculum stage to enhance the effectiveness of the joint framework. In particular, the proposed joint framework applied the proposed curriculum learning to a network large enough to learn various noise situations. Subsequently, the proposed ensemble distillation was applied to compress the large network into a network small enough to be onboard microcontrollers. The proposed joint framework achieved superior accuracy in noisy situations compared with state-of-the-art methods. In particular, the proposed joint framework achieved the small-footprint

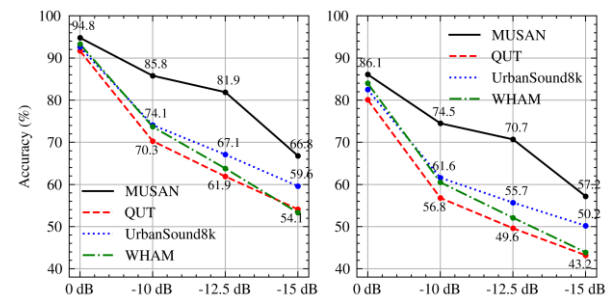


FIGURE 15. Accuracy of the (a) proposed joint framework and (b) curriculum learning according to the SNR on four noise datasets

keyword spotter with the accuracy of 79.1% at an SNR of -12.5 dB on the MUSAN dataset. Therefore, the proposed framework significantly improves the usability of voice interfaces in consumer electronics.

The limitations of our research and potential directions for future research are summarized as follows:

1) Our research targets low-performance, low-power, tiny consumer devices. The limitation of our research is that there needs to be more research on adaptive operation for devices with different performances and the number of microphones. Therefore, our future work is a joint framework considering hardware-specific conditions similar to the research field of hardware-aware neural architecture search [43].

2) We treated all sounds except voice commands as noise, but the audio-based consumer technology can advance to scene understanding through comprehensive noise analysis [30]. Recognizing the context of ambient noise can be used for better voice recognition or situational awareness. For example, the rate of cooking-related command recognition or service suggestion can be increased by recognizing a cooking situation with kitchen-related noise sound.

3) Although we showed the effect of the joint framework and presented a knowledge distillation method to enhance it, additional consideration is needed. Studies on new loss functions or knowledge transfer methods [44] have been actively conducted in knowledge distillation to boost teacher-student matching. Therefore, our future work will apply advanced loss functions and transfer methods to our joint framework to improve performance in loud-noise situations.

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